QUEZON CITY UNIVERSITY

San Bartolome, Novaliches, Quezon City
Bachelor of Science in Information Technology

RESEARCH PROPOSAL
MS102 – QUANTITATIVE METHODS

Group No.: 2 Section: SBIT-3B

Proposed Research Title: Impact of Generative AI Dependency on the Manual Coding Proficiency and Debugging Skills of 3rd Year BSIT Students		
Research Leader: Calderon, Anjelo S.		
Assistant Leader: Edica, Christian Jhuls B.		
Members:		
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2. Bautista, Winston M.	8. Malate, Sean T.	14. Rivera, Hamed R.
3. De Vera, Vaughn Ezekiel Reian	9. Nillar, Lee Bernard A	15. Rollan, Manolito Jr.
4. Fernandez, Sean M.	10. Osorio, Rheyvon E.	16. Sarmiento, Juvyen Lancelotte A.
5. Galope, Mark Christopher N.	11. Pahit, Noel M.	17. Sta. Ana, Banjo B.
6. Gardiola, Vincent	12. Parisan, John Rey D.	18. Tolentino, Daniel O.

I. Introduction

Background of Study

Programming education continues to evolve with the integration of artificial intelligence in software development. One emerging trend among students and developers is the heavy use of generative AI for code creation, where large portions of programs are produced using natural language prompts. Instead of manually designing and writing algorithms, users guide AI tools to generate code, rapidly prototype applications, and modify outputs based on the expected behavior of the program. This approach shifts development from structured manual coding to AI-assisted generation and iteration.

Third-year BSIT students handle advanced programming tasks that require strong independent competencies such as logical problem-solving, debugging, code comprehension, and program design. While generative AI enables faster development and convenience, its growing use raises concerns regarding students' ability to manually write, analyze, and correct programs on their own. Excessive dependency on AI-generated solutions may reduce opportunities for students to practice algorithmic thinking, debugging strategies, and structured coding techniques.

This study was conceived to examine how generative AI dependency influences the manual coding proficiency and debugging skills of third-year BSIT students. By understanding how AI-assisted coding behaviors relate to students' independent performance, the study may help guide learners and educators in using AI tools responsibly while preserving essential programming competencies.

Statement of the Problem

This study seeks to answer the following questions:

1. To what extent do third-year BSIT students depend on generative AI tools in programming tasks?
2. How does the level of generative AI dependency relate to manual programming competence in terms of:
   2.1 Writing code without AI assistance
   2.2 Debugging errors independently
   2.3 Solving programming problems lo gically
   2.4 Understanding and interpreting program code
3. What generative AI tools are commonly used by third-year BSIT students for programming activities?
4. Are there significant differences in manual coding proficiency and debugging skills between students with high and low levels of AI dependency?
5. What are the perceived advantages and challenges of using generative AI in programming activities according to students?

Hypothesis:
Generative AI dependency significantly affects the manual coding proficiency and debugging skills of third-year BSIT students.

Null Hypothesis:
Generative AI dependency has no significant effect on the manual coding proficiency and debugging skills of third-year BSIT students.

II. Purpose of the Study

This study is conducted to determine how dependency on generative AI affects the manual coding proficiency and debugging skills of third-year BSIT students. As AI-generated code and prompt-based development become increasingly common, there is a need to understand whether these practices enhance or weaken students' ability to write, analyze, and correct programs independently. The project seeks to address the problem of possible overreliance on AI tools that may limit logical reasoning, debugging strategies, and structured coding practice. By undertaking this study, students may become more aware of how their AI usage influences their learning, while instructors and the university may use the findings to improve teaching approaches and curriculum design in programming courses. If this study is not initiated, students may continue to depend heavily on AI-generated solutions without developing strong independent programming competence

Objectives

Main Objective:
To assess the impact of generative AI dependency on the manual coding proficiency and debugging skills of third-year BSIT students.

Specific Objectives:
1. To identify the extent to which third-year BSIT students depend on generative AI during programming tasks.
2. To measure manual coding proficiency and debugging skills in terms of code creation, problem-solving, debugging, and code comprehension.
3. To determine the common generative AI tools used by third-year BSIT students in programming activities.
4. To analyze the relationship between the level of AI dependency and manual coding proficiency and debugging skills.
5. To examine the perceived advantages and challenges of generative AI use as reported by students.

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Subject Adviser: Lalaine Josefa Carrao Date: _____

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RESEARCH PROPOSAL
MS102 – QUANTITATIVE METHODS

Group No.: 2 Section: SBIT-3B

Proposed Research Title: User Experience vs Integrity: Assessing the Usability and Perceived Efficacy of Anti-Cheat Mechanisms in LMS among TVL Students		
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I. Introduction

Background of Study

The increasing integration of digital platforms in secondary education has transformed the way instruction and assessment are delivered. Learning Management Systems (LMS) are now widely used to distribute learning materials, administer quizzes, and conduct online examinations. While these platforms provide flexibility and efficiency, they also raise concerns regarding academic integrity, particularly in preventing cheating during online assessments.

To address this issue, many LMS platforms implement anti-cheat mechanisms such as full-screen or kiosk modes, tab-switch detection, activity logging, time tracking, browser restrictions, and monitoring of student behavior during examinations. These features aim to reduce dishonest practices; however, they may also affect students' user experience. As security becomes stricter, usability problems such as anxiety, confusion, interruptions, and reduced ease of interaction may occur during assessments.

For TVL students, who require accessibility and efficiency in digital platforms, the balance between usability and integrity becomes critical. Overly restrictive systems may protect assessments but discourage effective interaction, while highly usable systems may lack sufficient protection against cheating.

This study is conducted to assess how anti-cheat mechanisms embedded in existing Learning Management Systems influence usability and perceived cheating prevention among TVL students. It focuses on students' perceptions and experiences with LMS security features and examines how these mechanisms affect comfort, interaction, and perceived fairness of online assessments.

Statement of the Problem

This study seeks to answer the following questions:

1. What is the level of usability of Learning Management Systems with anti-cheat mechanisms as perceived by TVL students in terms of:
   1.1 Ease of use
   1.2 System navigation
   1.3 Comfort during examinations
   1.4 Overall user experience
2. What is the level of perceived effectiveness of anti-cheat mechanisms in LMS in terms of:
   2.1 Cheating prevention
   2.2 Monitoring capability
   2.3 Fairness of online assessments
3. What challenges do TVL students experience when using LMS platforms with strict security controls during online examinations?
4. Is there a significant relationship between the perceived strictness of anti-cheat mechanisms and the usability of LMS platforms?
5. How do TVL students perceive the balance between usability and academic integrity in LMS-based online assessments?

Hypothesis

There is a significant relationship between the perceived usability of Learning Management Systems and the perceived effectiveness of anti-cheat mechanisms among TVL students.

Null Hypothesis

There is no significant relationship between the perceived usability of Learning Management Systems and the perceived effectiveness of anti-cheat mechanisms among TVL students.

II. Purpose of the Study

The purpose of this study is to assess the balance between user experience and academic integrity in Learning Management Systems that utilize anti-cheat mechanisms for online assessments. As digital examinations become more common, there is a growing need to understand how security features influence students' comfort, interaction, and perception of fairness while preventing dishonest practices. This study addresses whether stricter LMS security controls improve assessment integrity at the expense of usability. By evaluating students' experiences with existing LMS anti-cheat features, the research aims to identify how these mechanisms affect usability, stress, accessibility, and perceived effectiveness in cheating prevention. The findings of this study will benefit educators and institutions by providing evidence-based insights for selecting or configuring LMS platforms that maintain academic integrity without compromising student experience. Without this study, schools may continue implementing restrictive systems without understanding their impact on usability, learning efficiency, and student acceptance in online assessments.

Objectives

Main Objective:
To determine the relationship between perceived usability and perceived effectiveness of anti-cheat mechanisms in existing Learning Management Systems among TVL students.

Specific Objectives:
1. To evaluate the perceived usability of LMS platforms with anti-cheat mechanisms among TVL students.
2. To assess the perceived effectiveness of LMS anti-cheat features in preventing cheating during online examinations.
3. To identify common usability issues encountered by students when using secure LMS platforms.
4. To analyze the relationship between perceived strictness of security mechanisms and user experience.
5. To determine how usability and academic integrity can be balanced in LMS-based assessments.

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Subject Adviser: Lalaine Josefa Carrao Date: _____

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RESEARCH PROPOSAL
MS102 – QUANTITATIVE METHODS

Group No.: 2 Section: SBIT-3B

Proposed Research Title: Assessing Media and Information Literacy and Awareness of Digital Threats among College Students of Quezon City University SY 2025-2026		
Research Leader: Calderon, Anjelo S.		
Assistant Leader: Edica, Christian Jhuls B.		
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4. Fernandez, Sean M.	10. Osorio, Rheyvon E.	16. Sarmiento, Juvyen Lancelotte A.
5. Galope, Mark Christopher N.	11. Pahit, Noel M.	17. Sta. Ana, Banjo B.
6. Gardiola, Vincent	12. Parisan, John Rey D.	18. Tolentino, Daniel O.

I. Introduction
Background of the Study

In today's digital age, college students at Quezon City University increasingly rely on online platforms for learning, communication, research, and social interaction. While these tools provide efficiency and connectivity, they also expose students to numerous digital threats, such as phishing scams, malware, misinformation, and privacy breaches.

Media and Information Literacy (MIL), the ability to find, evaluate, and use digital information responsibly, plays a critical role in helping students navigate these risks safely. Students who lack MIL may fall victim to online scams, share personal data carelessly, or consume and spread inaccurate information. Conversely, students with strong MIL skills can make informed decisions, protect their digital footprint, and use technology more effectively.

Despite the importance of digital literacy, gaps remain among college students at Quezon City University. Some may have technical proficiency but lack awareness of online threats, while others may know about risks but struggle to evaluate information critically. Understanding these gaps is essential for designing programs, workshops, or courses that enhance students' knowledge and promote safer digital practices.

This study is conducted to assess the media and information literacy levels of college students at Quezon City University and examine how this literacy influences their awareness of digital threats and adoption of safe online practices. The research focuses on students' perceptions, knowledge, and self-reported behaviors, aiming to identify areas where interventions can improve digital literacy and security awareness.

Statement of the Problem

This study seeks to answer the following questions:

1. What is the level of media and information literacy among college students in terms of:
   1.1 Information evaluation and credibility assessment
   1.2 Awareness of digital threats (phishing, scams, malware, misinformation)
   1.3 Safe and responsible online behavior
   1.4 Cybersecurity practices (strong passwords, privacy settings, device security)
2. Does the level of media and information literacy influence students' awareness and behavior regarding digital threats?
3. Are there differences in MIL and digital threat awareness based on:
   3.1 Year level
   3.2 Program or strand
   3.3 Frequency of technology use
4. What challenges do students encounter due to limited media and information literacy?
5. Is there a significant relationship between media and information literacy and safe online practices among college students?

Hypothesis

Higher levels of media and information literacy among college students are associated with greater awareness of digital threats and safer online practices.

Null Hypothesis

Media and information literacy levels are not significantly associated with awareness of digital threats or online safety practices among college students.

II. Purpose of the Study

This study aims to assess the media and information literacy of college students at Quezon City University and examine how it influences their awareness of digital threats and adoption of safe online practices. Conducting this research is important because students, while frequently using digital tools for learning and communication, may lack sufficient knowledge to recognize and respond to online risks such as phishing, scams, malware, misinformation, or privacy breaches. For students, the study will provide insights into their own digital habits, helping them develop safer and more responsible online practices. For professors, it will highlight areas where students may require guidance or instruction on evaluating information critically and protecting their digital footprint. For the institution, the findings can inform the design of training programs, workshops, or policies to improve digital literacy and ensure a safer online learning environment. Without this study, students may continue to engage in unsafe digital behaviors, professors may be unaware of gaps in students' information-handling skills, and the institution may rely on assumed digital competence, leaving the university community vulnerable to online threats, misinformation, and potential academic and security risks.

Objectives

Main Objective

To determine the level of media and information literacy among college students and its relationship to awareness and safe practices regarding digital threats.

Specific Objectives

1. To assess students' competencies in finding, evaluating, and responsibly using digital information.
2. To evaluate students' awareness of common digital threats, including phishing, scams, malware, and misinformation.
3. To examine students' online behaviors and cybersecurity practices.
4. To analyze the relationship between MIL levels and awareness of digital threats.
5. To provide recommendations for interventions aimed at enhancing digital literacy and online safety among college students.

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Subject Adviser: Lalaine Josefa Carrao Date: _____

QUEZON CITY UNIVERSITY

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RESEARCH PROPOSAL
MS102 – QUANTITATIVE METHODS

Group No.: 2 Section: SBIT-3B

Proposed Research Title: Assessing the Impact of Digital Well-Being on Academic Performance and Stress Levels of College Students of Quezon City University SY 2025-2026		
Research Leader: Calderon, Anjelo S.		
Assistant Leader: Edica, Christian Jhuls B.		
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6. Gardiola, Vincent	12. Parisan, John Rey D.	18. Tolentino, Daniel O.

I. Introduction
Background of the Study

At Quezon City University, college students increasingly rely on digital devices and online platforms for studying, communication, and entertainment. While these tools provide convenience and access to information, many students struggle to manage their screen time effectively, which can lead to distractions, reduced focus, and decreased academic performance. Observations among students suggest that prolonged device use often contributes to mental fatigue and stress, especially during peak academic periods.

Digital well-being, which involves awareness and regulation of one's digital habits to maintain mental, emotional, and physical health, has become a growing concern for students. Despite the benefits of technology, students may not always recognize how excessive screen time or unregulated online engagement affects their learning, productivity, and overall well-being. This gap highlights the need to study their digital habits and identify patterns that may negatively influence academic outcomes.

This study was conceived to assess the digital well-being of college students at Quezon City University and explore how their digital habits impact stress levels and academic performance. The findings aim to provide insights that can guide interventions, awareness programs, or strategies to help students maintain a healthier balance between technology use and academic responsibilities, ultimately promoting better learning outcomes and overall student wellness.

Statement of the Problem

This study seeks to answer the following questions:

1. What are the daily digital habits of college students, including social media, gaming, and online learning usage?
2. What is the level of stress experienced by students due to digital device usage?
3. Is there a significant relationship between students' digital well-being and their academic performance?
4. How do students perceive the effects of excessive screen time on attention, productivity, and mental health?
5. What strategies do students employ to manage digital usage and maintain well-being?

Hypothesis

Higher levels of digital well-being among college students are associated with lower stress levels and better academic performance.

Null Hypothesis

Digital well-being levels have no significant relationship with stress levels or academic performance among college students.

II. Purpose of the Study

This study aims to assess the impact of digital well-being on the academic performance and stress levels of college students at Quezon City University. Conducting this research is important because, while digital devices and online platforms are essential for learning and communication, excessive or unregulated use may lead to stress, distraction, and decreased academic productivity. For students, the study will provide insights into their own digital habits, helping them develop strategies to manage screen time effectively, reduce stress, and improve focus and productivity. For professors, it will highlight how students' digital engagement affects learning outcomes, enabling them to design teaching approaches or provide guidance that supports healthier digital practices. For the institution, the findings can inform the creation of awareness campaigns, support programs, or policies to promote digital well-being across the university. Without this study, students may continue to experience digital fatigue, poor focus, and lower academic performance, professors

may remain unaware of the effects of students' digital habits on learning, and the institution may lack evidence-based strategies to support a safe, productive, and balanced digital environment.

Objectives

Main Objective
To determine the relationship between digital well-being, academic performance, and stress levels among college students.

Specific Objectives

1. To identify the daily digital habits of college students, including social media, gaming, and online learning usage.
2. To assess the level of stress experienced by students due to digital device usage.
3. To analyze the relationship between students' digital well-being and academic performance.
4. To examine students' perceptions of how excessive screen time affects attention, productivity, and mental health.
5. To provide recommendations for interventions, awareness programs, or strategies to improve digital well-being among college students.

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Subject Adviser: Lalaine Josefa Carrao Date: _____

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RESEARCH PROPOSAL
MS102 – QUANTITATIVE METHODS

Group No.: 2 Section: SBIT-3B

Proposed Research Title: The Effect of Procrastination on Academic Performance and Stress Levels among College Students of Quezon City University SY 2025-2026		
Research Leader: Calderon, Anjelo S.		
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6. Gardiola, Vincent	12. Parisan, John Rey D.	18. Tolentino, Daniel O.

I. Introduction
Background of the Study

Procrastination, or the tendency to delay tasks despite knowing the negative consequences, is a common challenge among college students. At Quezon City University, many students report postponing academic tasks such as assignments, projects, or exam preparation, which can lead to increased stress, last-minute work, and lower quality of outputs. While some students may manage their time effectively, others struggle to balance academic responsibilities with personal or extracurricular activities, which can affect learning outcomes and overall well-being.

This study was conceived to examine the prevalence of procrastination among college students, its impact on stress levels, and its relationship with academic performance. By understanding the factors that contribute to procrastination and its effects, this research aims to provide insights that can guide students, professors, and the university in developing strategies, interventions, or support systems to improve time management, reduce stress, and enhance academic success.

Statement of the Problem

This study seeks to answer the following questions:

1. What is the prevalence of procrastination among students in terms of delaying assignments, projects, and exam preparations?
2. What are the stress levels experienced by students as a result of procrastination?
3. How does procrastination affect students' academic performance, such as grades or GPA?
4. What are the common reasons or factors that contribute to procrastination among students?
5. Is there a significant relationship between procrastination, time management skills, and academic performance?
6. What strategies or practices do students currently use to cope with procrastination, and how effective are they perceived to be?

Hypothesis

Higher levels of procrastination among college students are associated with increased stress and lower academic performance.

Null Hypothesis

Procrastination levels have no significant relationship with stress or academic performance among college students.

II. Purpose of the Study

This study aims to investigate the impact of procrastination on academic performance and stress levels among college students at Quezon City University. Conducting this research is important because procrastination can lead to increased stress, decreased productivity, and poor academic outcomes. For students, the study will provide insights into their time management habits and help them identify strategies to overcome procrastination. For professors, it will highlight areas where students may require guidance or support in managing deadlines and academic responsibilities. For the institution, the findings can inform the development of workshops, awareness programs, or policies aimed at promoting effective time management and student well-being. Without this study, students may continue to struggle with delayed task completion, elevated stress, and lower academic performance, professors may be unaware of how procrastination affects learning outcomes, and the university may lack evidence-based strategies to support student productivity and success.

Objectives

Main Objective

To determine the relationship between digital well-being, academic performance, and stress levels among college students.

Specific Objectives

1. To assess the stress levels associated with procrastination.
2. To evaluate the impact of procrastination on academic performance.
3. To identify the factors contributing to procrastination among students.
4. To analyze the relationship between procrastination and time management skills.
5. To provide recommendations for strategies or interventions to reduce procrastination and improve academic outcomes.

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Subject Adviser: Lalaine Josefa Carrao Date: _____